

Claudio Landim: Measure Theory

Rigorous reference of definitions and statements

This document formalizes the previous course reference, with common notation, explicit hypotheses, and no proofs. The organization follows the 32 main lectures of the IMPA course. Coverage is grounded in the official lecture sequence and the complete annotated course guide listed at the end. Numbering here is internal to this reference, not the lecturer's numbering. Results previously marked as additions from the companion guide retain that designation. Unlisted auxiliary board lemmas and the three supplementary exercise-solution videos have not been exhaustively audited.

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Notation and conventions

The symbol Ω denotes a nonempty set. The sigma-algebra under discussion always consists of subsets of the stated underlying space.

Symbol	Definition or convention
$\mathbb{N}, \overline{\mathbb{R}}$	$\mathbb{N} = \{1, 2, \dots\}$ and $\overline{\mathbb{R}} = [-\infty, \infty]$.
$\mathcal{P}(\Omega)$	$\{A : A \subseteq \Omega\}$.
$A^c, A \triangle B$	$A^c = \Omega \setminus A$ and $A \triangle B = (A \setminus B) \cup (B \setminus A)$.
$\bigsqcup_j A_j$	$\bigcup_j A_j$, with the additional assertion $A_i \cap A_j = \emptyset$ for $i \neq j$.
$\mathbf{1}_A$	$\mathbf{1}_A(x) = 1$ if $x \in A$ and 0 if $x \notin A$.
$(\Omega, \mathcal{F}, \mu)$	$\mathcal{F}$ is a sigma-algebra on Ω and $\mu : \mathcal{F} \rightarrow [0, \infty]$ is a countably additive measure.
$\sigma(\mathcal{C}), \text{alg}(\mathcal{C})$	The sigma-algebra and algebra generated by $\mathcal{C} \subseteq \mathcal{P}(\Omega)$; see Section 2.
μ_0, μ^*	A premeasure and its induced outer measure.
$\overline{\mathcal{F}}^\mu, \bar{\mu}$	The completed sigma-algebra and measure.
$\mathcal{B}(\mathbb{R}^d), \mathcal{L}(\mathbb{R}^d), m_d$	The Borel sigma-algebra, its Lebesgue completion, and Lebesgue measure. Write $m = m_1$.
f^+, f^-	$f^+(x) = \max\{f(x), 0\}$ and $f^-(x) = \max\{-f(x), 0\}$. Thus $f = f^+ - f^-$ and $ f = f^+ + f^-$.
$[f], L^p(\mu)$	$[f] = \{g : g = f \text{ } \mu\text{-a.e.}\}$; L^p consists of these equivalence classes, not of distinguished pointwise representatives.

Almost everywhere. For a pointwise assertion $P(x)$, write

$$P(x) \text{ } \mu\text{-a.e.} \iff \exists N \in \mathcal{F} : \mu(N) = 0 \text{ and } \forall x \in \Omega \setminus N, P(x).$$

This definition does not require the exceptional set $\{x : \neg P(x)\}$ itself to be measurable. In particular, it applies on incomplete spaces.

Measure hypotheses. Unless explicitly specified, μ need not be finite, sigma-finite, or complete. The expressions *finite* and *sigma-finite* mean, respectively,

$$\mu(\Omega) < \infty, \quad \exists (E_n)_{n \in \mathbb{N}} \subseteq \mathcal{F} : \Omega = \bigcup_{n=1}^{\infty} E_n, \quad \forall n, \mu(E_n) < \infty.$$

A null set means an $N \in \mathcal{F}$ with $\mu(N) = 0$.

Borel restrictions. In comparisons such as $\mu_F \ll m$ or $\mu_F \perp m$, where μ_F is a Borel measure, m means $m|_{\mathcal{B}(\mathbb{R})}$, so the two measures have the same domain. The analogous restriction convention applies in $\mathbb{R}^d$.

Extended arithmetic. For nonnegative integration use $0 \cdot (+\infty) = 0$ and $\inf \emptyset = +\infty$. The expression $(+\infty) - (+\infty)$ is never assigned a value. A signed integral is used only if at least one of its positive and negative parts has finite integral. All L^p spaces and dual spaces in this document are over the real scalars.

Representatives. One may start with finite-a.e. measurable functions and replace their values on the measurable set where they are infinite by zero. Accordingly, L^0 and L^p classes will be represented by measurable functions $\Omega \rightarrow \mathbb{R}$. Equalities in L^p mean a.e. equalities. A pointwise statement about a representative explicitly identifies any exceptional null set that is needed.

1 Lecture 1: Nonmeasurable sets

Theorem 1.1 (Impossibility of measuring every subset of $\mathbb{R}$). *Assume the axiom of choice. There is no map $\lambda : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ satisfying all of*

$$\begin{aligned} \lambda(\emptyset) &= 0, \\ \forall a, b \in \mathbb{R}, a < b : \quad \lambda((a, b]) &= b - a, \\ \forall E \subseteq \mathbb{R} \quad \forall t \in \mathbb{R} : \quad \lambda(E + t) &= \lambda(E), \quad E + t = \{x + t : x \in E\}, \\ \forall (E_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}(\mathbb{R}) : \quad [\forall i \neq j, E_i \cap E_j &= \emptyset] \implies \lambda\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \lambda(E_n). \end{aligned}$$

Definition 1.2 (Vitali set). On $[0, 1]$ let $x \sim_{\mathbb{Q}} y$ iff $x - y \in \mathbb{Q}$. A Vitali set is a $V \subseteq [0, 1]$ such that

$$\forall x \in [0, 1] \quad \exists! v \in V : x - v \in \mathbb{Q}.$$

Theorem 1.3. *Under the axiom of choice a Vitali set exists, and every such set satisfies $V \notin \mathcal{L}(\mathbb{R})$.*

2 Lecture 2: Classes of sets and additive set functions

Definition 2.1 (Semi-algebra, algebra, sigma-algebra). Let $\mathcal{C} \subseteq \mathcal{P}(\Omega)$.

(i) $\mathcal{C}$ is a *semi-algebra* if

$$\emptyset, \Omega \in \mathcal{C}, \quad \forall A, B \in \mathcal{C}, \quad A \cap B \in \mathcal{C},$$

and

$$\forall A \in \mathcal{C} \quad \exists r \in \{0, 1, 2, \dots\} \quad \exists S_1, \dots, S_r \in \mathcal{C} : \quad \Omega \setminus A = \bigsqcup_{j=1}^r S_j.$$

The empty union is $\emptyset$.

(ii) $\mathcal{C}$ is an *algebra* if

$$\Omega \in \mathcal{C}, \quad \forall A \in \mathcal{C}, \quad A^c \in \mathcal{C}, \quad \forall A, B \in \mathcal{C}, \quad A \cup B \in \mathcal{C}.$$

(iii) $\mathcal{C}$ is a *sigma-algebra* if

$$\Omega \in \mathcal{C}, \quad \forall A \in \mathcal{C}, \quad A^c \in \mathcal{C}, \quad \forall (A_n)_{n \in \mathbb{N}} \subseteq \mathcal{C}, \quad \bigcup_{n=1}^{\infty} A_n \in \mathcal{C}.$$

An algebra is also closed under finite intersections and differences. A sigma-algebra is closed under countable intersections and differences.

Lemma 2.2 (Intersections and generated classes). *For every nonempty family $(\mathcal{A}_i)_{i \in I}$ of algebras, $\bigcap_{i \in I} \mathcal{A}_i$ is an algebra. The analogous assertion holds for sigma-algebras. Consequently, for $\mathcal{C} \subseteq \mathcal{P}(\Omega)$,*

$$\begin{aligned} \text{alg}(\mathcal{C}) &= \bigcap \{ \mathcal{A} : \mathcal{A} \text{ is an algebra on } \Omega, \mathcal{C} \subseteq \mathcal{A} \}, \\ \sigma(\mathcal{C}) &= \bigcap \{ \mathcal{F} : \mathcal{F} \text{ is a sigma-algebra on } \Omega, \mathcal{C} \subseteq \mathcal{F} \}. \end{aligned}$$

These are the unique smallest classes of the indicated types containing $\mathcal{C}$.

Lemma 2.3 (Algebra generated by a semi-algebra). *If $\mathcal{S}$ is a semi-algebra, then*

$$\text{alg}(\mathcal{S}) = \left\{ \bigsqcup_{j=1}^r S_j : r \in \{0, 1, 2, \dots\}, S_j \in \mathcal{S} \right\}.$$

Definition 2.4 (Additivity, premeasure, measure). Let $\emptyset \in \mathcal{C} \subseteq \mathcal{P}(\Omega)$ and let $\mu_0 : \mathcal{C} \rightarrow [0, \infty]$ satisfy $\mu_0(\emptyset) = 0$.

(i) μ_0 is *finitely additive* if for every $r \in \mathbb{N}$ and every disjoint $C_1, \dots, C_r \in \mathcal{C}$ with $\bigsqcup_{j=1}^r C_j \in \mathcal{C}$,

$$\mu_0 \left(\bigsqcup_{j=1}^r C_j \right) = \sum_{j=1}^r \mu_0(C_j).$$

(ii) μ_0 is *countably additive* if for every disjoint sequence $(C_n) \subseteq \mathcal{C}$ with $\bigsqcup_n C_n \in \mathcal{C}$,

$$\mu_0 \left(\bigsqcup_{n=1}^{\infty} C_n \right) = \sum_{n=1}^{\infty} \mu_0(C_n).$$

(iii) A *premeasure* is such a countably additive map on an algebra. A *measure* is such a countably additive map on a sigma-algebra.

Proposition 2.5 (Elementary identities and inequalities). *Let $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ be finitely additive on an algebra. For all $A, B \in \mathcal{A}$,*

$$A \subseteq B \implies \mu_0(A) \leq \mu_0(B), \quad \mu_0(A \cup B) \leq \mu_0(A) + \mu_0(B).$$

For $r \in \mathbb{N}$ and $A_1, \dots, A_r \in \mathcal{A}$, $\mu_0(\bigcup_{j=1}^r A_j) \leq \sum_{j=1}^r \mu_0(A_j)$. If $A \subseteq B$ and $\mu_0(B) < \infty$, then

$$\mu_0(B \setminus A) = \mu_0(B) - \mu_0(A).$$

For a measure these union inequalities also hold for countable families.

Example 2.6 (Counting, Dirac, and discrete measures). The following maps are measures on $\mathcal{P}(\Omega)$:

$$\mu_{\text{count}}(A) = \begin{cases} \#A, & A \text{ is finite,} \\ +\infty, & A \text{ is infinite,} \end{cases} \quad \delta_x(A) = \mathbf{1}_A(x) \quad (x \in \Omega).$$

For $(x_j)_{j \in \mathbb{N}} \subseteq \Omega$ and $(a_j)_{j \in \mathbb{N}} \subseteq [0, \infty)$,

$$\mu(A) = \sum_{j=1}^{\infty} a_j \mathbf{1}_A(x_j) \quad (A \subseteq \Omega)$$

is a measure. Repeated points are allowed.

Example 2.7 (Finite additivity does not imply countable additivity). Let $\Omega = (0, 1]$ and let $\mathcal{A}$ be the algebra of finite unions of intervals $(a, b]$, $0 \leq a < b \leq 1$, including $\emptyset$. Define

$$\mu_0(A) = \begin{cases} 1, & \exists \varepsilon > 0 : (0, \varepsilon] \subseteq A, \\ 0, & \text{otherwise.} \end{cases}$$

This map is finitely additive. However, the disjoint sets $B_n = (1/(n+1), 1/n]$ satisfy

$$\bigsqcup_{n=1}^{\infty} B_n = \Omega, \quad \mu_0(\Omega) = 1, \quad \sum_{n=1}^{\infty} \mu_0(B_n) = 0.$$

3 Lecture 3: Continuity and extension of set functions

Definition 3.1 (Monotone sequences of sets). For $A_n, A \subseteq \Omega$,

$$A_n \uparrow A \iff [\forall n, A_n \subseteq A_{n+1}] \text{ and } A = \bigcup_{n=1}^{\infty} A_n,$$

$$A_n \downarrow A \iff [\forall n, A_{n+1} \subseteq A_n] \text{ and } A = \bigcap_{n=1}^{\infty} A_n.$$

Theorem 3.2 (Continuity of a measure). For a measure μ and $(A_n) \subseteq \mathcal{F}$,

$$A_n \uparrow A \implies \lim_{n \rightarrow \infty} \mu(A_n) = \mu(A),$$

$$A_n \downarrow A, \quad \mu(A_1) < \infty \implies \lim_{n \rightarrow \infty} \mu(A_n) = \mu(A).$$

The first limit is increasing, and the second is decreasing.

Theorem 3.3 (Countable-additivity criteria). Let $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ be finitely additive with $\mu_0(\emptyset) = 0$. The following are equivalent:

- (i) μ_0 is a premeasure;
- (ii) for every $(A_n) \subseteq \mathcal{A}$ and $A \in \mathcal{A}$ with $A_n \uparrow A$, $\mu_0(A_n) \rightarrow \mu_0(A)$.

If additionally $\mu_0(\Omega) < \infty$, they are equivalent to

$$\forall (A_n) \subseteq \mathcal{A}, \quad A_n \downarrow \emptyset \implies \mu_0(A_n) \rightarrow 0.$$

Theorem 3.4 (Extension from a semi-algebra). Let $\mathcal{S}$ be a semi-algebra and let $\mu_0 : \mathcal{S} \rightarrow [0, \infty]$ be finitely additive with $\mu_0(\emptyset) = 0$. There exists exactly one finitely additive $\tilde{\mu}_0 : \text{alg}(\mathcal{S}) \rightarrow [0, \infty]$ extending μ_0 , and

$$\forall r \geq 0 \quad \forall \text{ disjoint } S_1, \dots, S_r \in \mathcal{S} : \quad \tilde{\mu}_0 \left(\bigsqcup_{j=1}^r S_j \right) = \sum_{j=1}^r \mu_0(S_j).$$

The right-hand side depends only on the union. If μ_0 is countably additive on $\mathcal{S}$, then $\tilde{\mu}_0$ is a premeasure on $\text{alg}(\mathcal{S})$.

4 Lecture 4: Outer measures and Carathéodory's theorem

Definition 4.1 (Outer measure). A map $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$ is an outer measure iff

$$\begin{aligned} \mu^*(\emptyset) &= 0, \\ \forall A, B \subseteq \Omega, \quad A \subseteq B &\implies \mu^*(A) \leq \mu^*(B), \\ \forall (A_n)_{n \in \mathbb{N}} \subseteq \mathcal{P}(\Omega), \quad \mu^* \left(\bigcup_{n=1}^{\infty} A_n \right) &\leq \sum_{n=1}^{\infty} \mu^*(A_n). \end{aligned}$$

Theorem 4.2 (Outer measure induced by a premeasure). If μ_0 is a premeasure on $\mathcal{A}$, then

$$\mu^*(E) = \inf \left\{ \sum_{n=1}^{\infty} \mu_0(A_n) : (A_n) \subseteq \mathcal{A}, E \subseteq \bigcup_{n=1}^{\infty} A_n \right\} \quad (E \subseteq \Omega)$$

is an outer measure and $\forall A \in \mathcal{A}, \mu^*(A) = \mu_0(A)$.

Definition 4.3 (Carathéodory measurability). For an outer measure μ^* , let

$$\mathcal{M}_{\mu^*} = \{E \subseteq \Omega : \forall T \subseteq \Omega, \quad \mu^*(T) = \mu^*(T \cap E) + \mu^*(T \setminus E)\}.$$

Members of $\mathcal{M}_{\mu^*}$ are called Carathéodory measurable.

Theorem 4.4 (Carathéodory). *For every outer measure μ^* , the class $\mathcal{M}_{\mu^*}$ is a sigma-algebra and $\mu^*|_{\mathcal{M}_{\mu^*}}$ is a complete measure. If μ^* is induced by a premeasure on $\mathcal{A}$, then*

$$\mathcal{A} \subseteq \sigma(\mathcal{A}) \subseteq \mathcal{M}_{\mu^*}.$$

Theorem 4.5 (Extension of a premeasure). *Every premeasure μ_0 on $\mathcal{A}$ has an extension to a measure on $\sigma(\mathcal{A})$, namely $\mu = \mu^*|_{\sigma(\mathcal{A})}$. If*

$$\exists (A_n) \subseteq \mathcal{A} : \quad \Omega = \bigcup_n A_n, \quad \forall n, \mu_0(A_n) < \infty,$$

then this extension is unique among measures on $\sigma(\mathcal{A})$.

Lemma 4.6 (Measurable envelopes; companion guide). *Let μ^* be induced by a premeasure on $\mathcal{A}$.*

- (i) *For every $E \subseteq \Omega$ there exists $B \in \sigma(\mathcal{A})$ with $E \subseteq B$ and $\mu^*(E) = \mu^*(B)$.*
- (ii) *Suppose $E \in \mathcal{M}_{\mu^*}$ and*

$$\exists (E_n) \subseteq \mathcal{M}_{\mu^*} : \quad E = \bigcup_n E_n, \quad \forall n, \mu^*(E_n) < \infty.$$

Then there exist $B_-, B_+ \in \sigma(\mathcal{A})$ with

$$B_- \subseteq E \subseteq B_+, \quad \mu^*(B_+ \setminus B_-) = 0.$$

5 Lecture 5: Monotone classes

Definition 5.1 (Monotone class). A class $\mathcal{H} \subseteq \mathcal{P}(\Omega)$ is monotone iff

$$\forall (H_n) \subseteq \mathcal{H} : \quad \begin{cases} H_n \subseteq H_{n+1} \quad \forall n \implies \bigcup_n H_n \in \mathcal{H}, \\ H_{n+1} \subseteq H_n \quad \forall n \implies \bigcap_n H_n \in \mathcal{H}. \end{cases}$$

For $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ define

$$\text{mon}(\mathcal{C}) = \bigcap \{ \mathcal{H} : \mathcal{H} \text{ is monotone, } \mathcal{C} \subseteq \mathcal{H} \}.$$

Theorem 5.2 (Monotone class theorem). *For every algebra $\mathcal{A}$ on Ω , $\text{mon}(\mathcal{A}) = \sigma(\mathcal{A})$.*

Corollary 5.3. *If $\mathcal{H}$ is monotone and $\mathcal{A} \subseteq \mathcal{H}$ for an algebra $\mathcal{A}$, then $\sigma(\mathcal{A}) \subseteq \mathcal{H}$. An algebra is a sigma-algebra iff it is also a monotone class.*

Corollary 5.4 (Uniqueness of measures). *Let μ, ν be measures on $\sigma(\mathcal{A})$ such that $\forall A \in \mathcal{A}, \mu(A) = \nu(A)$. If both measures are finite, then $\mu = \nu$ on $\sigma(\mathcal{A})$. More generally, the same conclusion holds if*

$$\exists (A_n) \subseteq \mathcal{A} : \quad \Omega = \bigcup_n A_n, \quad \forall n, \mu(A_n) = \nu(A_n) < \infty.$$

6 Lecture 6: Construction of Lebesgue measure

Definition 6.1 (Interval semi-algebra and length). On $\mathbb{R}$, let

$$\mathcal{S}_{\text{int}} = \{\emptyset\} \cup \{(a, b] \cap \mathbb{R} : a, b \in \overline{\mathbb{R}}, a < b\}.$$

Here $(a, +\infty] \cap \mathbb{R} = (a, +\infty)$ and $(-\infty, b] \cap \mathbb{R} = (-\infty, b]$. Define

$$m_0(\emptyset) = 0, \quad m_0((a, b]) = b - a \quad (a, b \in \mathbb{R}, a < b),$$

and assign $+\infty$ to every nonempty unbounded member.

Theorem 6.2 (Lebesgue measure). *The class $\mathcal{S}_{\text{int}}$ is a semi-algebra, $\sigma(\mathcal{S}_{\text{int}}) = \mathcal{B}(\mathbb{R})$, and m_0 is countably additive on $\mathcal{S}_{\text{int}}$. There is exactly one Borel measure m with*

$$\forall a, b \in \mathbb{R}, a < b : \quad m((a, b]) = b - a.$$

Its completion has domain $\mathcal{L}(\mathbb{R})$ and is also denoted m .

Proposition 6.3 (Open-interval formula for outer measure). *For every $E \subseteq \mathbb{R}$,*

$$m^*(E) = \inf \left\{ \sum_{n=1}^{\infty} (b_n - a_n) : a_n, b_n \in \mathbb{R}, a_n \leq b_n, E \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n) \right\}.$$

Degenerate intervals $(a_n, a_n) = \emptyset$ are permitted. The Lebesgue sigma-algebra is $\mathcal{L}(\mathbb{R}) = \mathcal{M}_{m^}$.*

7 Lecture 7: Identities and invariance of Lebesgue measure

Proposition 7.1 (Null points and intervals). *For every $x \in \mathbb{R}$, $m(\{x\}) = 0$. For every countable $E \subseteq \mathbb{R}$, $E \in \mathcal{L}(\mathbb{R})$ and $m(E) = 0$. For every $a, b \in \mathbb{R}$ with $a \leq b$,*

$$m((a, b)) = m((a, b]) = m([a, b)) = m([a, b]) = b - a.$$

Theorem 7.2 (Translation and scaling). *For $E \subseteq \mathbb{R}$, $t \in \mathbb{R}$, and $c \in \mathbb{R} \setminus \{0\}$, define $E + t = \{x + t : x \in E\}$ and $cE = \{cx : x \in E\}$. Then*

$$m^*(E + t) = m^*(E), \quad m^*(cE) = |c|m^*(E),$$

and

$$E \in \mathcal{L}(\mathbb{R}) \iff E + t \in \mathcal{L}(\mathbb{R}) \iff cE \in \mathcal{L}(\mathbb{R}).$$

For such measurable E , the same identities hold with m in place of m^ .*

Corollary 7.3 (Finite-dimensional version). *For $d \in \mathbb{N}$ there is a unique Borel measure m_d on $\mathbb{R}^d$ with*

$$m_d \left(\prod_{j=1}^d (a_j, b_j] \right) = \prod_{j=1}^d (b_j - a_j) \quad (a_j, b_j \in \mathbb{R}, a_j < b_j).$$

Write $\mathcal{L}(\mathbb{R}^d)$ for its completion and again m_d for the completed measure. Its associated outer measure is

$$m_d^*(E) = \inf \{m_d(B) : B \in \mathcal{B}(\mathbb{R}^d), E \subseteq B\} \quad (E \subseteq \mathbb{R}^d).$$

For every invertible linear map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $t \in \mathbb{R}^d$,

$$\forall E \subseteq \mathbb{R}^d : \quad m_d^*(TE + t) = |\det T| m_d^*(E),$$

$$E \in \mathcal{L}(\mathbb{R}^d) \iff TE + t \in \mathcal{L}(\mathbb{R}^d), \quad m_d(TE + t) = |\det T| m_d(E) \quad (E \in \mathcal{L}(\mathbb{R}^d)).$$

8 Lecture 8: Complete measures

Definition 8.1 (Complete measure space). $(\Omega, \mathcal{F}, \mu)$ is complete iff

$$\forall Z \in \mathcal{F} \forall N \subseteq Z : \mu(Z) = 0 \implies N \in \mathcal{F}.$$

Monotonicity then gives $\mu(N) = 0$.

Theorem 8.2 (Completion). *For every measure space define*

$$\overline{\mathcal{F}}^\mu = \{A \cup N : A \in \mathcal{F}, \exists Z \in \mathcal{F}, N \subseteq Z, \mu(Z) = 0\}.$$

This is a sigma-algebra and $\bar{\mu}(A \cup N) = \mu(A)$ is a well-defined complete measure extending μ . Equivalently,

$$E \in \overline{\mathcal{F}}^\mu \iff \exists A, B \in \mathcal{F} : A \subseteq E \subseteq B, \mu(B \setminus A) = 0.$$

For such a sandwich, $\bar{\mu}(E) = \mu(A) = \mu(B)$. Every complete measure on $\overline{\mathcal{F}}^\mu$ extending μ equals $\bar{\mu}$. Every complete extension on a larger sigma-algebra contains $\overline{\mathcal{F}}^\mu$ and restricts to $\bar{\mu}$ on it.

Corollary 8.3. *If μ_0 is a sigma-finite premeasure, μ^* its induced outer measure, and $\mu = \mu^*|_{\sigma(\mathcal{A})}$, then*

$$\mathcal{M}_{\mu^*} = \overline{\sigma(\mathcal{A})}^\mu, \quad \mu^*|_{\mathcal{M}_{\mu^*}} = \bar{\mu}.$$

In particular, $\mathcal{L}(\mathbb{R}^d) = \overline{\mathcal{B}(\mathbb{R}^d)}^{m_d}$.

9 Lecture 9: Approximation of measurable sets

Definition 9.1 (Regularity in the course formulation). Let X have topology τ , let $\mathcal{B}(X) = \sigma(\tau) \subseteq \mathcal{F}$, and let μ be a measure on $\mathcal{F}$. The closed-open approximation property is

$$\forall E \in \mathcal{F} \forall \varepsilon > 0 \quad \exists C \subseteq X \text{ closed } \exists U \in \tau : C \subseteq E \subseteq U, \mu(U \setminus C) < \varepsilon.$$

This is the regularity formulation used here. The Borel measure is $\mu_B = \mu|_{\mathcal{B}(X)}$.

Proposition 9.2. *Under this approximation property, $\mathcal{F} \subseteq \overline{\mathcal{B}(X)}^{\mu_B}$. More explicitly, for every $E \in \mathcal{F}$ there are C_n closed and U_n open such that, with $C = \bigcup_n C_n$ and $G = \bigcap_n U_n$,*

$$C \subseteq E \subseteq G, \quad \mu(G \setminus C) = 0.$$

Theorem 9.3 (Approximation by the generating algebra). *Let μ be the extension of a premeasure on $\mathcal{A}$ furnished by Carathéodory's construction, either on $\sigma(\mathcal{A})$ or on its completion. Then*

$$\forall E \text{ in this domain with } \mu(E) < \infty \forall \varepsilon > 0 \quad \exists A \in \mathcal{A} : \mu(E \Delta A) < \varepsilon.$$

Theorem 9.4 (Lebesgue regularity). *For every $d \in \mathbb{N}$ and $E \in \mathcal{L}(\mathbb{R}^d)$,*

$$m_d(E) = \inf\{m_d(U) : E \subseteq U, U \text{ open}\} = \sup\{m_d(K) : K \subseteq E, K \text{ compact}\}.$$

Moreover,

$$\forall \varepsilon > 0 \quad \exists C \text{ closed } \exists U \text{ open} : C \subseteq E \subseteq U, m_d(U \setminus C) < \varepsilon.$$

If $m_d(E) < \infty$, C may be chosen compact.

Corollary 9.5 (Exact approximation modulo null sets). *For every $E \in \mathcal{L}(\mathbb{R}^d)$ there exist closed sets (C_n) and open sets (U_n) with*

$$\bigcup_n C_n \subseteq E \subseteq \bigcap_n U_n, \quad m_d\left(\left(\bigcap_n U_n\right) \setminus \left(\bigcup_n C_n\right)\right) = 0.$$

A set of the form $\bigcup_n C_n$ is called F_σ ; a set of the form $\bigcap_n U_n$ is called G_δ .

10 Lecture 10: Measurable maps and simple functions

Definition 10.1 (Measurable map). For measurable spaces $(X, \mathcal{F})$ and $(Y, \mathcal{G})$, $f : X \rightarrow Y$ is $(\mathcal{F}, \mathcal{G})$ -measurable iff

$$\forall B \in \mathcal{G}, \quad f^{-1}(B) = \{x \in X : f(x) \in B\} \in \mathcal{F}.$$

For $\mathbb{R}$ or $\overline{\mathbb{R}}$ the target sigma-algebra is its Borel sigma-algebra.

Lemma 10.2 (Generating-class criterion). If $\mathcal{G} = \sigma(\mathcal{C})$ for $\mathcal{C} \subseteq \mathcal{P}(Y)$, then

$$f \text{ is } (\mathcal{F}, \mathcal{G})\text{-measurable} \iff \forall C \in \mathcal{C}, \quad f^{-1}(C) \in \mathcal{F}.$$

Proposition 10.3 (Threshold characterizations). For $f : \Omega \rightarrow \overline{\mathbb{R}}$, the following are equivalent:

$$f \text{ measurable}; \quad \forall a \in \mathbb{R}, \quad \{f > a\} \in \mathcal{F}; \quad \forall a \in \mathbb{R}, \quad \{f \geq a\} \in \mathcal{F};$$

$$\forall a \in \mathbb{R}, \quad \{f < a\} \in \mathcal{F}; \quad \forall a \in \mathbb{R}, \quad \{f \leq a\} \in \mathcal{F}.$$

In each threshold condition, $\mathbb{R}$ may be replaced by $\mathbb{Q}$. Here $\{f > a\} = \{x : f(x) > a\}$, and similarly for the other comparisons. If these hold, $\{f = +\infty\}, \{f = -\infty\} \in \mathcal{F}$.

Definition 10.4 (Simple function and its integral). A function $s : \Omega \rightarrow \mathbb{R}$ is simple iff it is measurable and its range $s(\Omega)$ is finite. Equivalently, there exist $r \in \mathbb{N}$, $a_1, \dots, a_r \in \mathbb{R}$, and $A_1, \dots, A_r \in \mathcal{F}$ with

$$s = \sum_{j=1}^r a_j \mathbf{1}_{A_j}.$$

If $a_1, \dots, a_r$ are the distinct values of s , then $A_j = \{s = a_j\}$ gives a disjoint partition of Ω . For a nonnegative simple function define, using this partition,

$$I_\mu(s) = \int s \, d\mu = \sum_{j=1}^r a_j \mu(A_j).$$

Lemma 10.5 (Independence of the representation). If $s = \sum_{j=1}^r a_j \mathbf{1}_{A_j}$ with $a_j \geq 0$ and $A_j \in \mathcal{F}$, then $I_\mu(s) = \sum_{j=1}^r a_j \mu(A_j)$, even if the A_j overlap. In particular, every nonnegative-coefficient representation gives the same value.

11 Lecture 11: Closure properties of measurable functions

Proposition 11.1 (Algebraic closure). If $f, g : \Omega \rightarrow \mathbb{R}$ are measurable, then

$$f + g, \quad f - g, \quad fg, \quad |f|, \quad \max\{f, g\}, \quad \min\{f, g\}$$

are measurable. So is the function

$$h(x) = \begin{cases} f(x)/g(x), & g(x) \neq 0, \\ 0, & g(x) = 0. \end{cases}$$

Consequently f/g is measurable whenever g is everywhere nonzero. The corresponding extended-real operations are measurable on their measurable domains of definition.

Proposition 11.2 (Countable limits and extrema). *If each $f_n : \Omega \rightarrow \overline{\mathbb{R}}$ is measurable, then so are*

$$\sup_n f_n, \quad \inf_n f_n, \quad \limsup_{n \rightarrow \infty} f_n = \inf_{k \in \mathbb{N}} \sup_{n \geq k} f_n, \quad \liminf_{n \rightarrow \infty} f_n = \sup_{k \in \mathbb{N}} \inf_{n \geq k} f_n.$$

The set $E = \{x : \liminf_n f_n(x) = \limsup_n f_n(x)\}$ is measurable. The pointwise limit on E , extended by zero to $\Omega \setminus E$, is measurable. If $f_n \rightarrow g$ μ -a.e., this construction gives a measurable f with $f = g$ μ -a.e., even if the specified g is not measurable.

Proposition 11.3 (Composition and continuity). *If $f : (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $h : (Y, \mathcal{G}) \rightarrow (Z, \mathcal{H})$ are measurable, then $h \circ f : (X, \mathcal{F}) \rightarrow (Z, \mathcal{H})$ is measurable. If $h : X \rightarrow Y$ is continuous between topological spaces, then it is $(\mathcal{B}(X), \mathcal{B}(Y))$ -measurable.*

Example 11.4 (Uncountable suprema). Let $V \subseteq \mathbb{R}$ be a non-Lebesgue-measurable set. Every $f_v = \mathbf{1}_{\{v\}}$, $v \in V$, is Borel measurable, but

$$\sup_{v \in V} f_v = \mathbf{1}_V$$

is not Lebesgue measurable.

Proposition 11.5 (Modification on a null set). *If $(\Omega, \mathcal{F}, \mu)$ is complete, $f : \Omega \rightarrow \overline{\mathbb{R}}$ is measurable, and $g : \Omega \rightarrow \overline{\mathbb{R}}$ satisfies $g = f$ μ -a.e., then g is measurable.*

Proposition 11.6 (Representatives for the completion; companion guide). *If $f : \Omega \rightarrow \overline{\mathbb{R}}$ is $\overline{\mathcal{F}}^\mu$ -measurable, there exists an $\mathcal{F}$ -measurable $g : \Omega \rightarrow \overline{\mathbb{R}}$ with $f = g$ $\bar{\mu}$ -a.e. In particular, every Lebesgue-measurable function on $\mathbb{R}^d$ has a Borel-measurable representative equal to it m_d -a.e.*

12 Lecture 12: Simple approximation and the integral

Theorem 12.1 (Increasing simple approximation). *If $f : \Omega \rightarrow [0, \infty]$ is measurable, define for $n \in \mathbb{N}$*

$$s_n(x) = 2^{-n} \lfloor 2^n \min\{f(x), n\} \rfloor.$$

Every s_n is a nonnegative simple function, $0 \leq s_n \leq s_{n+1} \leq f$, and $\forall x \in \Omega$, $s_n(x) \rightarrow f(x)$. If $f(x) \leq n$, then $0 \leq f(x) - s_n(x) < 2^{-n}$. If $\sup_{x \in \Omega} f(x) < \infty$, then

$$\forall \varepsilon > 0 \exists n_0 \in \mathbb{N} \forall n \geq n_0 \forall x \in \Omega : |f(x) - s_n(x)| < \varepsilon.$$

Definition 12.2 (Nonnegative integral). For a measurable $f : \Omega \rightarrow [0, \infty]$, define

$$\int f d\mu = \sup\{I_\mu(s) : s \text{ nonnegative simple, } \forall x, s(x) \leq f(x)\} \in [0, \infty].$$

Lemma 12.3 (Comparison with simple approximations). *Let $0 \leq s_n \uparrow f$ pointwise, with all s_n simple. For every nonnegative simple $\varphi \leq f$, $I_\mu(\varphi) \leq \lim_n I_\mu(s_n)$. Consequently,*

$$\int f d\mu = \lim_{n \rightarrow \infty} \int s_n d\mu$$

for every such increasing simple approximation, not just the dyadic one.

Definition 12.4 (Signed integral and integrability). For measurable $f : \Omega \rightarrow \overline{\mathbb{R}}$, let $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$. If $\min\{\int f^+ d\mu, \int f^- d\mu\} < \infty$, define

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu.$$

If both terms are infinite the signed integral is undefined. The function f is integrable iff

$$\int |f| d\mu < \infty \iff \int f^+ d\mu < \infty \text{ and } \int f^- d\mu < \infty.$$

13 Lecture 13: Integration of simple functions

Proposition 13.1. *For nonnegative simple functions s, t and $c \in [0, \infty)$,*

$$I_\mu(s + t) = I_\mu(s) + I_\mu(t), \quad I_\mu(cs) = cI_\mu(s), \quad s \leq t \implies I_\mu(s) \leq I_\mu(t).$$

For an integrable real simple $s = \sum_{j=1}^r a_j \mathbf{1}_{A_j}$ with disjoint level sets, $\int s d\mu = \sum_{j=1}^r a_j \mu(A_j)$. In this finite sum $0 \cdot \infty = 0$; every nonzero term is finite. The integrable real simple functions form a real vector space, and

$$\forall a, b \in \mathbb{R} : \quad \int (as + bt) d\mu = a \int s d\mu + b \int t d\mu$$

for integrable simple s, t .

14 Lecture 14: Algebraic and order properties of the integral

Theorem 14.1 (Nonnegative additivity and homogeneity). *If $f, g : \Omega \rightarrow [0, \infty]$ are measurable and $c \in [0, \infty)$, then*

$$\int (f + g) d\mu = \int f d\mu + \int g d\mu, \quad \int cf d\mu = c \int f d\mu.$$

Values may be $+\infty$.

Theorem 14.2 (Linearity). *Let $\mathcal{L}^1(\mu) = \{f : \Omega \rightarrow \mathbb{R} : f \text{ measurable, } \int |f| d\mu < \infty\}$. This is a real vector space and, for $f, g \in \mathcal{L}^1(\mu)$ and $a, b \in \mathbb{R}$,*

$$\int (af + bg) d\mu = a \int f d\mu + b \int g d\mu.$$

Integration descends to a linear functional on $L^1(\mu) = \mathcal{L}^1(\mu)/\sim$, where $f \sim g$ iff $f = g$ μ -a.e..

Proposition 14.3 (Order and the integral triangle inequality). *If $f, g : \Omega \rightarrow \overline{\mathbb{R}}$ are measurable, both signed integrals are defined, and $f \leq g$ μ -a.e., then $\int f d\mu \leq \int g d\mu$. For integrable f ,*

$$\left| \int f d\mu \right| \leq \int |f| d\mu.$$

Proposition 14.4 (Domination). *If $f : \Omega \rightarrow \overline{\mathbb{R}}$ is measurable and there is a nonnegative integrable g with $|f| \leq g$ μ -a.e., then f is integrable. If $f \in L^1(\mu)$ and h is measurable with $|h| \leq M$ μ -a.e. for some finite $M \geq 0$, then*

$$hf \in L^1(\mu), \quad \int |hf| d\mu \leq M \int |f| d\mu.$$

15 Lecture 15: Null sets and restriction of integrals

Proposition 15.1 (Finiteness and zero integral). *For measurable $f : \Omega \rightarrow \overline{\mathbb{R}}$,*

$$\int |f| d\mu < \infty \implies \mu(\{|f| = \infty\}) = 0.$$

For measurable $f : \Omega \rightarrow [0, \infty]$,

$$\int f d\mu = 0 \iff \exists N \in \mathcal{F} : \mu(N) = 0, \forall x \notin N, f(x) = 0.$$

Proposition 15.2 (A.e. invariance). *Let $f, g : \Omega \rightarrow \overline{\mathbb{R}}$ be measurable and $f = g$ μ -a.e.. Then $\int f^\pm d\mu = \int g^\pm d\mu$. Thus the signed integrals are defined simultaneously and are equal whenever defined. For $N \in \mathcal{F}$ with $\mu(N) = 0$ and measurable $u \geq 0$, $\int u \mathbf{1}_N d\mu = 0$, including when u is infinite on N .*

Definition 15.3 (Integral over a measurable set). For $A \in \mathcal{F}$ set

$$f \mathbf{1}_A(x) = \begin{cases} f(x), & x \in A, \\ 0, & x \notin A, \end{cases} \quad \int_A f d\mu = \int f \mathbf{1}_A d\mu$$

whenever the right side is defined.

Proposition 15.4 (Restriction and countable disjoint unions). *If $f \in L^1(\mu)$ and $A \in \mathcal{F}$, then $f \mathbf{1}_A \in L^1(\mu)$. If $(A_j)_{j \in \mathbb{N}} \subseteq \mathcal{F}$ is disjoint, then*

$$\int_{\bigsqcup_j A_j} f d\mu = \sum_{j=1}^{\infty} \int_{A_j} f d\mu$$

for every measurable $f \geq 0$ and for every integrable real f . In the latter case,

$$\sum_{j=1}^{\infty} \left| \int_{A_j} f d\mu \right| \leq \int_{\bigcup_j A_j} |f| d\mu < \infty.$$

16 Lecture 16: Convergence theorems for integrals

Theorem 16.1 (Monotone convergence). *Let $f_n, f : \Omega \rightarrow [0, \infty]$ be measurable. If there is $N \in \mathcal{F}$ with $\mu(N) = 0$ such that*

$$\forall x \notin N : \quad 0 \leq f_n(x) \leq f_{n+1}(x) \quad (\forall n), \quad \lim_n f_n(x) = f(x),$$

then $\int f_n d\mu \uparrow \int f d\mu$. More generally, the same conclusion holds for measurable $f_n, f : \Omega \rightarrow \overline{\mathbb{R}}$ when $f_n \uparrow f$ μ -a.e. and there is an integrable h with $f_n \geq h$ μ -a.e. for every n . All those integrals are then defined in $(-\infty, \infty]$.

Corollary 16.2 (Nonnegative series). *For every sequence of measurable $u_n : \Omega \rightarrow [0, \infty]$,*

$$\int \left(\sum_{n=1}^{\infty} u_n \right) d\mu = \sum_{n=1}^{\infty} \int u_n d\mu.$$

Theorem 16.3 (Fatou's lemma and its signed variants). *If $f_n : \Omega \rightarrow [0, \infty]$ are measurable, then*

$$\int \liminf_n f_n d\mu \leq \liminf_n \int f_n d\mu.$$

The same inequality holds for measurable $f_n : \Omega \rightarrow \overline{\mathbb{R}}$ if $f_n \geq h$ μ -a.e. for every n , with h integrable. If instead $f_n \leq g$ μ -a.e. for every n , with g integrable, then

$$\limsup_n \int f_n d\mu \leq \int \limsup_n f_n d\mu.$$

In the lower-bound case the integrals have no value $-\infty$; in the upper-bound case they have no value $+\infty$.

Theorem 16.4 (Dominated convergence). *Let $f_n, f : \Omega \rightarrow \overline{\mathbb{R}}$ be measurable. Suppose $f_n \rightarrow f$ μ -a.e. and there is a nonnegative integrable g with $|f_n| \leq g$ μ -a.e. for every n . Then every f_n and f is integrable, and*

$$\lim_n \int |f_n - f| d\mu = 0, \quad \lim_n \int f_n d\mu = \int f d\mu.$$

Differences may be evaluated after choosing finite real representatives on the common null set.

Proposition 16.5 (Measure defined by a density). *For measurable $w : \Omega \rightarrow [0, \infty]$, define $\mu_w(A) = \int_A w d\mu$ for $A \in \mathcal{F}$. Then μ_w is a measure and $\forall A \in \mathcal{F}, \mu(A) = 0 \implies \mu_w(A) = 0$.*

Proposition 16.6 (Absolute continuity of an integral). *For every integrable f ,*

$$\forall \varepsilon > 0 \exists \delta > 0 \forall A \in \mathcal{F} : \mu(A) < \delta \implies \int_A |f| d\mu < \varepsilon.$$

Corollary 16.7 (Continuity of parameter integrals; companion guide). *Let $I \subseteq \mathbb{R}$ be an open interval and $H : I \times \Omega \rightarrow \mathbb{R}$. Suppose*

- (i) $H(t, \cdot)$ is measurable for every $t \in I$;
- (ii) there is $N \in \mathcal{F}, \mu(N) = 0$, such that $t \mapsto H(t, x)$ is continuous on I for every $x \notin N$;
- (iii) there is nonnegative $g \in L^1(\mu)$ such that $\forall x \notin N \forall t \in I, |H(t, x)| \leq g(x)$.

Then $J(t) = \int H(t, x) d\mu(x)$ is finite and continuous on I .

Corollary 16.8 (Differentiation under the integral; companion guide). *Let $I \subseteq \mathbb{R}$ be an open interval and $H : I \times \Omega \rightarrow \mathbb{R}$. Suppose $H(t, \cdot)$ is integrable for every $t \in I$. Suppose there exist a measurable null N and nonnegative $g \in L^1(\mu)$ such that*

$$\forall x \notin N : H(\cdot, x) \text{ is differentiable on } I, \quad \forall t \in I : |\partial_t H(t, x)| \leq g(x).$$

For each t , extend $\partial_t H(t, \cdot)$ by zero on N . It is measurable and integrable, and

$$J(t) = \int H(t, x) d\mu(x) \implies \forall t \in I, J'(t) = \int \partial_t H(t, x) d\mu(x).$$

17 Lecture 17: Products of measure spaces

Definition 17.1 (Product sigma-algebra, rectangles, and sections). *For measurable spaces $(X, \mathcal{F}_1)$ and $(Y, \mathcal{F}_2)$ define*

$$\mathcal{F}_1 \otimes \mathcal{F}_2 = \sigma\{A \times B : A \in \mathcal{F}_1, B \in \mathcal{F}_2\}.$$

The collection of measurable rectangles $\{A \times B : A \in \mathcal{F}_1, B \in \mathcal{F}_2\}$ is a semi-algebra on $X \times Y$. For $E \subseteq X \times Y$ define

$$E_x = \{y \in Y : (x, y) \in E\}, \quad E^y = \{x \in X : (x, y) \in E\}.$$

For $f : X \times Y \rightarrow \overline{\mathbb{R}}$ write $f_x(y) = f(x, y)$ and $f^y(x) = f(x, y)$.

Lemma 17.2 (Measurability of sections). *For every $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$, $E_x \in \mathcal{F}_2$ for every $x \in X$ and $E^y \in \mathcal{F}_1$ for every $y \in Y$. If f is product measurable, every f_x and every f^y is measurable.*

Theorem 17.3 (Product measure). *Let $(X, \mathcal{F}_1, \mu_1)$ and $(Y, \mathcal{F}_2, \mu_2)$ be sigma-finite measure spaces. There is exactly one measure $\mu_1 \otimes \mu_2$ on $\mathcal{F}_1 \otimes \mathcal{F}_2$ satisfying*

$$\forall A \in \mathcal{F}_1 \forall B \in \mathcal{F}_2 : (\mu_1 \otimes \mu_2)(A \times B) = \mu_1(A)\mu_2(B).$$

This measure is sigma-finite. The formula uses $0 \cdot \infty = 0$. For every finite family of sigma-finite spaces the iterated product measure is independent of the parenthesization, and

$$\left(\bigotimes_{j=1}^r \mu_j \right) \left(\prod_{j=1}^r A_j \right) = \prod_{j=1}^r \mu_j(A_j) \quad (A_j \in \mathcal{F}_j).$$

Proposition 17.4 (Borel products; companion guide). *If X and Y are separable metric spaces and $X \times Y$ carries the product topology, then $\mathcal{B}(X \times Y) = \mathcal{B}(X) \otimes \mathcal{B}(Y)$.*

Example 17.5 (An incomplete product). Take the complete spaces $(\mathbb{R}, \mathcal{L}(\mathbb{R}), m)$ in both factors. The product-measurable set $\{0\} \times \mathbb{R}$ has product measure zero. If $V \subseteq \mathbb{R}$ is not Lebesgue measurable, then

$$\{0\} \times V \subseteq \{0\} \times \mathbb{R}, \quad \{0\} \times V \notin \mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R}).$$

The product space is therefore incomplete.

18 Lecture 18: Countable products of probability spaces

Definition 18.1 (Probability space, cylinders, independence). A measure space $(\Omega, \mathcal{F}, \mu)$ is a probability space iff $\mu(\Omega) = 1$. For a sequence of probability spaces $(\Omega_j, \mathcal{F}_j, \mu_j)$, let

$$\Omega = \prod_{j=1}^{\infty} \Omega_j, \quad \pi_j((x_i)_{i \in \mathbb{N}}) = x_j, \quad \mathcal{F} = \sigma\{\pi_j^{-1}(A) : j \in \mathbb{N}, A \in \mathcal{F}_j\}.$$

A rectangle cylinder is a set

$$C(A_1, \dots, A_r) = \bigcap_{j=1}^r \pi_j^{-1}(A_j) = \prod_{j=1}^r A_j \times \prod_{j>r} \Omega_j, \quad A_j \in \mathcal{F}_j.$$

For a probability μ on $(\Omega, \mathcal{F})$, the coordinates are independent with distributions μ_j iff

$$\forall r \in \mathbb{N} \forall \text{ distinct } j_1, \dots, j_r \forall A_i \in \mathcal{F}_{j_i} : \mu \left(\bigcap_{i=1}^r \pi_{j_i}^{-1}(A_i) \right) = \prod_{i=1}^r \mu_{j_i}(A_i).$$

In particular, the distribution of π_j is the pushforward $(\pi_j)_*\mu$, defined by $(\pi_j)_*\mu(A) = \mu(\pi_j^{-1}(A))$.

Theorem 18.2 (Countable independent product). *The rectangle cylinders form a semi-algebra. The assignment*

$$\mu_0(C(A_1, \dots, A_r)) = \prod_{j=1}^r \mu_j(A_j)$$

is well defined and extends to a premeasure on their generated algebra. There is a unique probability measure $\mu = \bigotimes_{j=1}^{\infty} \mu_j$ on $\mathcal{F}$ extending it. Under μ the coordinates are independent with distributions μ_j . No standard Borel assumption is needed for this independent-product theorem.

19 Lecture 19: Tonelli's and Fubini's theorems

Throughout this section the two factor measures μ_1, μ_2 are sigma-finite, and functions on $X \times Y$ are $\mathcal{F}_1 \otimes \mathcal{F}_2$ -measurable.

Lemma 19.1 (Measurability of section integrals). *If $f : X \times Y \rightarrow [0, \infty]$ is measurable, then*

$$x \mapsto \int_Y f(x, y) d\mu_2(y), \quad y \mapsto \int_X f(x, y) d\mu_1(x)$$

are measurable extended-nonnegative functions. For every $E \in \mathcal{F}_1 \otimes \mathcal{F}_2$,

$$(\mu_1 \otimes \mu_2)(E) = \int_X \mu_2(E_x) d\mu_1(x) = \int_Y \mu_1(E^y) d\mu_2(y).$$

Theorem 19.2 (Tonelli). *For every measurable $f : X \times Y \rightarrow [0, \infty]$,*

$$\begin{aligned} \int_{X \times Y} f d(\mu_1 \otimes \mu_2) &= \int_X \left(\int_Y f(x, y) d\mu_2(y) \right) d\mu_1(x) \\ &= \int_Y \left(\int_X f(x, y) d\mu_1(x) \right) d\mu_2(y). \end{aligned}$$

All three values belong to $[0, \infty]$.

Theorem 19.3 (Fubini). *Let $f : X \times Y \rightarrow \mathbb{R}$ be measurable and $\int |f| d(\mu_1 \otimes \mu_2) < \infty$. The sets*

$$X_0 = \left\{ x : \int_Y |f(x, y)| d\mu_2(y) < \infty \right\}, \quad Y_0 = \left\{ y : \int_X |f(x, y)| d\mu_1(x) < \infty \right\}$$

are measurable and satisfy $\mu_1(X \setminus X_0) = \mu_2(Y \setminus Y_0) = 0$. Define

$$G(x) = \begin{cases} \int_Y f(x, y) d\mu_2(y), & x \in X_0, \\ 0, & x \notin X_0, \end{cases} \quad H(y) = \begin{cases} \int_X f(x, y) d\mu_1(x), & y \in Y_0, \\ 0, & y \notin Y_0. \end{cases}$$

Then $G \in L^1(\mu_1)$, $H \in L^1(\mu_2)$, and

$$\int f d(\mu_1 \otimes \mu_2) = \int_X G d\mu_1 = \int_Y H d\mu_2.$$

Corollary 19.4 (Absolute-integrability criterion). *For measurable $f : X \times Y \rightarrow \mathbb{R}$, each of the following is equivalent:*

$$\begin{aligned} f \in L^1(\mu_1 \otimes \mu_2), \quad \int_X \int_Y |f(x, y)| d\mu_2(y) d\mu_1(x) &< \infty, \\ \int_Y \int_X |f(x, y)| d\mu_1(x) d\mu_2(y) &< \infty. \end{aligned}$$

20 Lecture 20: Signed measures, Hahn and Jordan decompositions

Definition 20.1 (Signed measure). A signed measure is a map $\nu : \mathcal{F} \rightarrow [-\infty, \infty]$ such that

- (i) $\nu(\emptyset) = 0$;
- (ii) it does not assume both $+\infty$ and $-\infty$ as values;

(iii) for every disjoint sequence $(A_n) \subseteq \mathcal{F}$, the series $\sum_n \nu(A_n)$ is defined and $\nu(\bigsqcup_n A_n) = \sum_n \nu(A_n)$.

For extended real numbers a_n , the series is defined by $\sum_n a_n = \sum_n a_n^+ - \sum_n a_n^-$ when at least one of the two nonnegative sums is finite. A signed measure is called finite if $|\nu|(\Omega) < \infty$, where total variation is defined below.

Lemma 20.2 (Additivity and continuity). *For $A, B \in \mathcal{F}$ with $A \subseteq B$, $\nu(B) = \nu(A) + \nu(B \setminus A)$. If $\nu(B) \in \mathbb{R}$, both terms on the right are finite. For $(A_n) \subseteq \mathcal{F}$,*

$$A_n \uparrow A \implies \nu(A_n) \rightarrow \nu(A), \quad A_n \downarrow A, \nu(A_1) \in \mathbb{R} \implies \nu(A_n) \rightarrow \nu(A).$$

Definition 20.3 (Positive and negative sets). For a signed measure ν and $P, N \in \mathcal{F}$,

$$P \text{ positive} \iff \forall A \in \mathcal{F}, A \subseteq P \implies \nu(A) \geq 0,$$

$$N \text{ negative} \iff \forall A \in \mathcal{F}, A \subseteq N \implies \nu(A) \leq 0.$$

A Hahn decomposition is a pair (P, N) of such sets with $P \cap N = \emptyset$ and $P \cup N = \Omega$.

Lemma 20.4 (Minimum-set lemma). *If $\nu(A) > -\infty$ for every $A \in \mathcal{F}$, then*

$$\beta = \inf_{A \in \mathcal{F}} \nu(A) > -\infty, \quad \exists N \in \mathcal{F} : \nu(N) = \beta.$$

Theorem 20.5 (Hahn decomposition). *Every signed measure has a Hahn decomposition.*

Theorem 20.6 (Jordan decomposition). *For any Hahn decomposition (P, N) set*

$$\nu^+(A) = \nu(A \cap P), \quad \nu^-(A) = -\nu(A \cap N) \quad (A \in \mathcal{F}).$$

These are nonnegative measures, they satisfy

$$\forall A \in \mathcal{F}, \quad \nu(A) = \nu^+(A) - \nu^-(A), \quad \min\{\nu^+(\Omega), \nu^-(\Omega)\} < \infty,$$

and they are mutually singular: there is $S \in \mathcal{F}$ with $\nu^+(S^c) = 0$ and $\nu^-(S) = 0$. They are independent of the Hahn decomposition and are the unique mutually singular nonnegative measures with this difference. For two Hahn decompositions (P, N) and (P', N') , $|\nu|(P \triangle P') = |\nu|(N \triangle N') = 0$.

Definition 20.7 (Total variation). For a signed measure define the nonnegative measure $|\nu| = \nu^+ + \nu^-$. For $A \in \mathcal{F}$ it also satisfies

$$|\nu|(A) = \sup \left\{ \sum_{j=1}^r |\nu(A_j)| : r \in \mathbb{N}, A_j \in \mathcal{F}, A = \bigsqcup_{j=1}^r A_j \right\}.$$

Proposition 20.8 (Null sets and finiteness). *For every $A \in \mathcal{F}$,*

$$|\nu|(A) = 0 \iff \forall B \in \mathcal{F}, B \subseteq A \implies \nu(B) = 0.$$

Moreover,

$$[\forall A \in \mathcal{F}, \nu(A) \in \mathbb{R}] \iff |\nu|(\Omega) < \infty.$$

21 Lecture 21: Absolute continuity and measure decompositions

Definition 21.1 (Absolute continuity and singularity). For nonnegative measures μ, ν on the same $(\Omega, \mathcal{F})$,

$$\nu \ll \mu \iff \forall A \in \mathcal{F}, \mu(A) = 0 \implies \nu(A) = 0,$$

$$\nu \perp \mu \iff \exists S \in \mathcal{F} : \mu(S) = 0, \quad \nu(S^c) = 0.$$

For signed ν , define $\nu \ll \mu$ by $|\nu| \ll \mu$ and $\nu \perp \mu$ by $|\nu| \perp \mu$. For two signed measures, singularity means singularity of their total variations. For a signed ν , $\nu \ll \mu$ is also equivalent to $\forall A \in \mathcal{F}, \mu(A) = 0 \implies \nu(A) = 0$.

Proposition 21.2. *Absolute continuity is reflexive and transitive: $\mu \ll \mu$, and $\lambda \ll \nu \ll \mu \implies \lambda \ll \mu$. Singularity is symmetric: $\nu \perp \mu \iff \mu \perp \nu$.*

Theorem 21.3 (Radon–Nikodym). *Let μ, ν be sigma-finite nonnegative measures on $(\Omega, \mathcal{F})$. If $\nu \ll \mu$, there is a measurable $h : \Omega \rightarrow [0, \infty]$ with*

$$\forall A \in \mathcal{F}, \quad \nu(A) = \int_A h \, d\mu.$$

Any two such densities are equal μ -a.e., and $h < \infty$ μ -a.e.. Its equivalence class is denoted $d\nu/d\mu$. If ν is finite signed, μ is sigma-finite, and $\nu \ll \mu$, there is a unique $h \in L^1(\mu)$ with the same formula.

Theorem 21.4 (Lebesgue decomposition). *For sigma-finite nonnegative μ, ν on $(\Omega, \mathcal{F})$ there are unique nonnegative measures ν_{ac}, ν_s with*

$$\nu = \nu_{ac} + \nu_s, \quad \nu_{ac} \ll \mu, \quad \nu_s \perp \mu.$$

Both parts are sigma-finite and $\nu_{ac}(A) = \int_A h \, d\mu$ for a unique a.e. nonnegative density h . For finite signed ν and sigma-finite μ , the same assertion holds with finite signed parts and $h \in L^1(\mu)$.

Corollary 21.5 (Extended signed version). *Let ν be a signed measure with $|\nu|$ sigma-finite and let μ be sigma-finite. Apply the preceding theorem to ν^+ and ν^- :*

$$\nu^\pm = h_\pm \mu + \rho_\pm, \quad \rho_\pm \perp \mu.$$

Then

$$\nu_{ac} = (h_+ - h_-)\mu, \quad \nu_s = \rho_+ - \rho_-, \quad \nu = \nu_{ac} + \nu_s$$

is the unique signed decomposition with the stated absolute continuity and singularity properties. Densities may be chosen finite off one μ -null set, and at least one of $\int h_+ \, d\mu, \int h_- \, d\mu$ is finite. Thus all displayed signed integrals exclude $\infty - \infty$.

Lemma 21.6 (Uniqueness from integrals on sets). *For $f, g \in L^1(\mu)$,*

$$\left[\forall A \in \mathcal{F}, \int_A f \, d\mu = \int_A g \, d\mu \right] \iff f = g \quad \mu\text{-a.e..}$$

Proposition 21.7 (Integration with a density). *Let $h : \Omega \rightarrow [0, \infty]$ be measurable and $\nu(A) = \int_A h \, d\mu$. For every measurable $u : \Omega \rightarrow [0, \infty]$,*

$$\int u \, d\nu = \int u h \, d\mu.$$

For measurable real u , $u \in L^1(\nu) \iff \int |u| h \, d\mu < \infty$; under this condition the same signed identity holds. Products with zero use $0 \cdot \infty = 0$.

22 Lecture 22: Almost everywhere and almost uniform convergence

Definition 22.1 (Equivalence classes and a.e. convergence). For measurable $f, g : \Omega \rightarrow \mathbb{R}$, define

$$f \sim g \iff f = g \quad \mu\text{-a.e.}, \quad L^0(\mu) = \{f : \Omega \rightarrow \mathbb{R} \text{ measurable}\} / \sim.$$

For real measurable f_n, f , a.e. convergence means

$$f_n \rightarrow f \quad \mu\text{-a.e.} \iff \exists N \in \mathcal{F}, \mu(N) = 0, \quad \forall x \notin N \quad \forall \varepsilon > 0 \\ \exists n_0 = n_0(x, \varepsilon) \in \mathbb{N} \quad \forall n \geq n_0 : \quad |f_n(x) - f(x)| < \varepsilon.$$

Definition 22.2 (Essential bound and L^∞). For measurable real f , define

$$\|f\|_\infty = \inf\{M \in [0, \infty) : \mu(\{|f| > M\}) = 0\} \in [0, \infty].$$

For $u \geq 0$, $\text{ess sup}_\mu u$ denotes this same bound on u . Let

$$\mathcal{L}^\infty(\mu) = \{f : \Omega \rightarrow \mathbb{R} \text{ measurable} : \|f\|_\infty < \infty\}, \quad L^\infty(\mu) = \mathcal{L}^\infty(\mu) / \sim.$$

The value is unchanged on an equivalence class and is a norm on $L^\infty(\mu)$.

Definition 22.3 (Uniform convergence outside a null set). For measurable real f_n, f , this convergence means

$$\exists N \in \mathcal{F}, \mu(N) = 0, \quad \forall \varepsilon > 0 \quad \exists n_0 \in \mathbb{N} \quad \forall n \geq n_0 \quad \forall x \notin N : \\ |f_n(x) - f(x)| < \varepsilon.$$

It is equivalent to $\text{ess sup}_\mu |f_n - f| \rightarrow 0$. If $f_n, f \in L^\infty$, this is precisely convergence in its norm.

Definition 22.4 (Almost uniform convergence). Write $f_n \rightarrow f$ almost uniformly iff

$$\forall \eta > 0 \quad \exists E \in \mathcal{F}, \mu(E) < \eta, \quad \forall \varepsilon > 0 \quad \exists n_0 \in \mathbb{N} \\ \forall n \geq n_0 \quad \forall x \in \Omega \setminus E : \quad |f_n(x) - f(x)| < \varepsilon.$$

The set E may depend on η but not on ε or n .

Theorem 22.5 (Egorov). If $\mu(\Omega) < \infty$ and real measurable f_n, f satisfy $f_n \rightarrow f$ μ -a.e., then $f_n \rightarrow f$ almost uniformly.

Proposition 22.6 (Implications and uniqueness). On every measure space,

$$\text{uniformly outside a null set} \implies \text{almost uniformly} \implies \mu\text{-a.e.}$$

Almost uniform convergence also implies convergence in measure (defined in the next section). On a finite measure space, a.e. convergence is equivalent to almost uniform convergence. Each limit under these modes is unique up to a.e. equality.

23 Lecture 23: Convergence in measure

Definition 23.1 (Convergence and Cauchy sequences in measure). For measurable $f_n, f : \Omega \rightarrow \mathbb{R}$, define

$$f_n \xrightarrow{\mu} f \iff \forall \varepsilon > 0 \quad \forall \eta > 0 \quad \exists n_0 \in \mathbb{N} \quad \forall n \geq n_0 : \\ \mu(\{x : |f_n(x) - f(x)| > \varepsilon\}) < \eta.$$

The sequence is Cauchy in measure iff

$$\forall \varepsilon > 0 \quad \forall \eta > 0 \quad \exists n_0 \in \mathbb{N} \quad \forall n, m \geq n_0 : \quad \mu(\{|f_n - f_m| > \varepsilon\}) < \eta.$$

These definitions are invariant under a.e. changes of representatives.

Theorem 23.2 (Subsequences and uniqueness). *If $f_n \xrightarrow{\mu} f$, there is a strictly increasing sequence $(n_k)_{k \in \mathbb{N}}$ with $f_{n_k} \rightarrow f$ μ -a.e.. If $f_n \xrightarrow{\mu} f$ and $f_n \xrightarrow{\mu} g$, then $f = g$ μ -a.e..*

Theorem 23.3 (Finite-measure implication). *If $\mu(\Omega) < \infty$ and $f_n \rightarrow f$ μ -a.e., then $f_n \xrightarrow{\mu} f$.*

Example 23.4 (A.e. convergence without convergence in measure). On $(\mathbb{R}, \mathcal{L}(\mathbb{R}), m)$ let $f_n = \mathbf{1}_{[n, n+1]}$. For every $x \in \mathbb{R}$, $f_n(x) \rightarrow 0$, but

$$\forall n \in \mathbb{N} \forall \varepsilon \in (0, 1) : m(\{|f_n| > \varepsilon\}) = 1.$$

Example 23.5 (Convergence in measure without a.e. convergence). On $[0, 1)$ with Lebesgue measure, for integers $k \geq 0$ and $0 \leq j < 2^k$ define

$$f_{2^k+j} = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k})}.$$

For $0 < \varepsilon < 1$, $m(\{|f_{2^k+j}| > \varepsilon\}) = 2^{-k}$, so $f_n \xrightarrow{m} 0$. For every $x \in [0, 1)$, $\limsup_n f_n(x) = 1$ and $\liminf_n f_n(x) = 0$.

Theorem 23.6 (Completeness in measure; companion guide). *On every measure space, every sequence of measurable real functions that is Cauchy in measure has a measurable real limit in measure. If $\mu(\Omega) < \infty$, the formula*

$$d_0([f], [g]) = \int \min\{1, |f - g|\} d\mu$$

is a metric on $L^0(\mu)$, and $d_0([f_n], [f]) \rightarrow 0 \iff f_n \xrightarrow{\mu} f$. The metric space $(L^0(\mu), d_0)$ is complete.

24 Lecture 24: Young, Hölder, and Minkowski inequalities

The norms and equivalence-class spaces used below are formally defined in Section 25. For $1 \leq p < \infty$, $\|f\|_p = (f |f|^p d\mu)^{1/p}$; for $p = \infty$ use Section 22. Two exponents $p, q \in [1, \infty]$ are conjugate iff $1/p + 1/q = 1$, with $1/\infty = 0$.

Theorem 24.1 (Young's inequality). *If $1 < p < \infty$ and $q = p/(p-1)$, then*

$$\forall a, b \in [0, \infty) : ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Definition 24.2 (Convex conjugate). A proper convex function $\Phi : \mathbb{R} \rightarrow \mathbb{R} \cup \{+\infty\}$ has nonempty effective domain $\{a : \Phi(a) < \infty\}$ and satisfies

$$\forall a, c \in \mathbb{R} \forall t \in (0, 1) : \Phi(ta + (1-t)c) \leq t\Phi(a) + (1-t)\Phi(c).$$

Its convex conjugate is $\Phi^*(b) = \sup_{a \in \mathbb{R}} \{ab - \Phi(a)\}$.

Proposition 24.3 (Fenchel–Young form). *For every such Φ and all $a, b \in \mathbb{R}$, $ab \leq \Phi(a) + \Phi^*(b)$. For $\Phi(a) = |a|^p/p$, $1 < p < \infty$, $\Phi^*(b) = |b|^q/q$. Equivalently, the conjugate of a^p/p on $a \geq 0$, evaluated at $b \geq 0$, is b^q/q .*

Theorem 24.4 (Hölder). *For conjugate $p, q \in [1, \infty]$ and $f \in L^p(\mu)$, $g \in L^q(\mu)$,*

$$fg \in L^1(\mu), \quad \int |fg| d\mu \leq \|f\|_p \|g\|_q.$$

This includes $(p, q) = (1, \infty)$ and $(\infty, 1)$. For $p = q = 2$ it yields Cauchy–Schwarz: $|\int fg d\mu| \leq \|f\|_2 \|g\|_2$.

Theorem 24.5 (Minkowski). *For every $p \in [1, \infty]$ and $f, g \in L^p(\mu)$, $f + g \in L^p(\mu)$ and $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.*

25 Lecture 25: The spaces L^p and completeness

Definition 25.1 (L^p and its norm). For $1 \leq p < \infty$ let

$$\mathcal{L}^p(\mu) = \left\{ f : \Omega \rightarrow \mathbb{R} : f \text{ measurable, } \int |f|^p d\mu < \infty \right\}, \quad L^p(\mu) = \mathcal{L}^p(\mu)/\sim.$$

For $[f] \in L^p(\mu)$ define $\|[f]\|_p = (\int |f|^p d\mu)^{1/p}$. For $p = \infty$ use the definition in Section 22. Each L^p is a real vector space with $[f] + [g] = [f + g]$ and $a[f] = [af]$; its norm is independent of the representative. One customarily writes f in place of $[f]$.

Definition 25.2 (Norm convergence, Cauchy sequence, Banach space). A real normed space is a real vector space V with a map $\|\cdot\| : V \rightarrow [0, \infty)$ satisfying, for all $u, v \in V$ and $c \in \mathbb{R}$,

$$\|v\| = 0 \iff v = 0, \quad \|cv\| = |c| \|v\|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

In such a space,

$$v_n \rightarrow v \text{ in norm} \iff \forall \varepsilon > 0 \exists n_0 \forall n \geq n_0 : \|v_n - v\| < \varepsilon.$$

A sequence is norm-Cauchy iff $\forall \varepsilon > 0 \exists n_0 \forall n, m \geq n_0, \|v_n - v_m\| < \varepsilon$. A Banach space is a normed space in which every norm-Cauchy sequence has a norm limit in the space.

Theorem 25.3 (Completeness of L^p). For every measure space and every $p \in [1, \infty]$, $L^p(\mu)$ is a Banach space.

Proposition 25.4 (Chebyshev's inequality and convergence consequences). For $1 \leq p < \infty$, $f, g \in L^p(\mu)$, and $\varepsilon > 0$,

$$\mu(\{|f - g| > \varepsilon\}) \leq \varepsilon^{-p} \|f - g\|_p^p.$$

For every $p \in [1, \infty]$, $|\|f\|_p - \|g\|_p| \leq \|f - g\|_p$. Thus $f_n \rightarrow f$ in L^p implies convergence of norms. For finite p it also implies convergence in measure and a.e. convergence of a subsequence. For $p = \infty$ it implies uniform convergence outside a single null set.

Lemma 25.5 (Sigma-finite support). For $f \in \mathcal{L}^p(\mu)$, $1 \leq p < \infty$, let $S_f = \{x : |f(x)| > 0\}$. Then

$$S_f = \bigcup_{n=1}^{\infty} \{|f| \geq 1/n\}, \quad \mu(\{|f| \geq 1/n\}) \leq n^p \|f\|_p^p < \infty.$$

Hence the restriction of μ to S_f is sigma-finite.

Corollary 25.6 (Embedding for finite measure; companion guide). If $0 < \mu(\Omega) < \infty$ and $1 \leq p < q \leq \infty$, then

$$L^q(\mu) \subseteq L^p(\mu), \quad \forall f \in L^q(\mu), \quad \|f\|_p \leq \mu(\Omega)^{1/p-1/q} \|f\|_q.$$

If $\mu(\Omega) = 0$, all these spaces are the zero vector space.

Corollary 25.7 (Hilbert structure; companion guide). The formula $\langle [f], [g] \rangle = \int fg d\mu$ is a well-defined real inner product on $L^2(\mu)$: for all $f, g, h \in L^2(\mu)$ and $a, b \in \mathbb{R}$,

$$\langle af + bg, h \rangle = a\langle f, h \rangle + b\langle g, h \rangle, \quad \langle f, g \rangle = \langle g, f \rangle,$$

$$\langle f, f \rangle \geq 0, \quad \langle f, f \rangle = 0 \iff f = 0 \text{ in } L^2(\mu).$$

Its induced norm is $\|f\|_2 = \sqrt{\langle f, f \rangle}$. The space is complete for this norm, and hence is a Hilbert space.

26 Lecture 26: From convergence in measure to L^p convergence

Definition 26.1 (Equicontinuity at the empty set). For a family $(\nu_i)_{i \in I}$ of finite nonnegative measures on $\mathcal{F}$, equicontinuity at $\emptyset$ means

$$\forall (E_k)_{k \in \mathbb{N}} \subseteq \mathcal{F}, \quad E_k \downarrow \emptyset \implies \lim_{k \rightarrow \infty} \sup_{i \in I} \nu_i(E_k) = 0.$$

Definition 26.2 (Uniform absolute continuity). The family $(\nu_i)_{i \in I}$ is uniformly absolutely continuous with respect to μ iff

$$\forall \varepsilon > 0 \exists \delta > 0 \forall A \in \mathcal{F} : \quad \mu(A) < \delta \implies \sup_{i \in I} \nu_i(A) < \varepsilon.$$

Lemma 26.3. If $(\nu_i)_{i \in I}$ is equicontinuous at $\emptyset$ and $\forall i \in I, \nu_i \ll \mu$, then it is uniformly absolutely continuous with respect to μ . This assertion does not require μ to be finite or sigma-finite.

Theorem 26.4 (Global L^p convergence criterion). Fix $1 \leq p < \infty$, let $f_n \in \mathcal{L}^p(\mu)$, and let $f : \Omega \rightarrow \mathbb{R}$ be measurable. Define finite measures

$$\nu_n(A) = \int_A |f_n|^p d\mu \quad (A \in \mathcal{F}).$$

Then

$$\begin{aligned} f \in \mathcal{L}^p(\mu) \text{ and } \|f_n - f\|_p \rightarrow 0 \\ \iff \left[f_n \xrightarrow{\mu} f \text{ and } \forall (E_k) \subseteq \mathcal{F}, E_k \downarrow \emptyset \implies \sup_n \int_{E_k} |f_n|^p d\mu \rightarrow 0 \right]. \end{aligned}$$

There is no sigma-finiteness assumption on μ .

Corollary 26.5 (Domination in L^p). If $f_n \xrightarrow{\mu} f$ and there is a nonnegative $g \in L^p(\mu)$ with $|f_n| \leq g$ μ -a.e. for every n , then $f \in L^p(\mu)$ and $\|f_n - f\|_p \rightarrow 0$.

Corollary 26.6 (Finite-measure criterion). If $\mu(\Omega) < \infty$, $f_n \in L^p(\mu)$, $f_n \xrightarrow{\mu} f$, and the measures $\nu_n = |f_n|^p \mu$ are uniformly absolutely continuous with respect to μ , then $f \in L^p(\mu)$ and $\|f_n - f\|_p \rightarrow 0$.

Corollary 26.7 (Convergence of norms). For $1 \leq p < \infty$ and $f_n, f \in L^p(\mu)$,

$$f_n \xrightarrow{\mu} f, \quad \|f_n\|_p \rightarrow \|f\|_p \implies \|f_n - f\|_p \rightarrow 0.$$

Definition 26.8 (Uniform integrability on a finite measure space). For $\mu(\Omega) < \infty$, a family $(u_i)_{i \in I} \subseteq L^1(\mu)$ is uniformly integrable iff

$$\lim_{R \rightarrow \infty} \sup_{i \in I} \int_{\{|u_i| > R\}} |u_i| d\mu = 0.$$

Theorem 26.9 (Uniform integrability criterion). If $\mu(\Omega) < \infty$, $1 \leq p < \infty$, $f_n \in L^p(\mu)$, and $f : \Omega \rightarrow \mathbb{R}$ is measurable, then

$$f \in L^p(\mu), \quad \|f_n - f\|_p \rightarrow 0 \iff f_n \xrightarrow{\mu} f, \quad \lim_{R \rightarrow \infty} \sup_n \int_{\{|f_n|^p > R\}} |f_n|^p d\mu = 0.$$

Example 26.10 (Escape of mass on an infinite measure space). On $\mathbb{R}$, $f_n = \mathbf{1}_{[n, n+1]}$ satisfies

$$\forall R > 1 : \quad \sup_n \int_{\{|f_n|^p > R\}} |f_n|^p dm = 0, \quad \|f_n\|_p = 1.$$

Thus a tail bound alone does not force L^p convergence to the a.e. limit 0. It also does not replace the global convergence-in-measure hypothesis. More directly, $g_n = n^{-1/p} \mathbf{1}_{[0, n]}$ satisfies

$$g_n \xrightarrow{m} 0, \quad \sup_n \int_{\{|g_n|^p > R\}} |g_n|^p dm = 0 \quad (R > 1), \quad \|g_n\|_p = 1.$$

For $E_k = [k, \infty)$, $\sup_n \int_{E_k} |g_n|^p dm = 1$, while $E_k \downarrow \emptyset$.

27 Lecture 27: Bounded linear functionals and the dual of L^p

Definition 27.1 (Bounded linear functional and dual norm). For a real normed space V , a map $T : V \rightarrow \mathbb{R}$ is linear iff

$$\forall u, v \in V \quad \forall a, b \in \mathbb{R} : \quad T(au + bv) = aT(u) + bT(v).$$

It is bounded iff $\exists C < \infty \quad \forall v \in V, \quad |T(v)| \leq C \|v\|$. The dual space and its norm are

$$V^* = \{T : V \rightarrow \mathbb{R} : T \text{ linear and bounded}\}, \quad \|T\| = \sup_{\|v\| \leq 1} |T(v)|.$$

A linear functional is bounded iff it is continuous for the norm topology. Equivalently, continuity at zero means $\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall v, \quad \|v\| < \delta \implies |T(v)| < \varepsilon$.

Proposition 27.2 (Integration functionals). For conjugate $p, q \in [1, \infty]$ and $g \in L^q(\mu)$, define

$$T_g : L^p(\mu) \rightarrow \mathbb{R}, \quad T_g(f) = \int fg \, d\mu.$$

It is a well-defined bounded linear functional and $\|T_g\| \leq \|g\|_q$. If μ is sigma-finite, equality holds.

Theorem 27.3 (Dual representation). Assume μ is sigma-finite, $1 \leq p < \infty$, and $1/p + 1/q = 1$. For every $T \in (L^p(\mu))^*$ there exists a unique $g \in L^q(\mu)$ with

$$\forall f \in L^p(\mu), \quad T(f) = \int fg \, d\mu, \quad \|T\| = \|g\|_q.$$

Thus $g \mapsto T_g$ is a linear isometric bijection $L^q(\mu) \rightarrow (L^p(\mu))^*$. For $1 < p < \infty$ the same theorem holds on arbitrary measure spaces. For $p = 1$ the stated sigma-finite version gives $(L^1(\mu))^* \cong L^\infty(\mu)$.

Proposition 27.4 (The endpoint $p = \infty$). The map $J : L^1(\mu) \rightarrow (L^\infty(\mu))^*$, $J(g) = T_g$, is a linear isometric injection: $\|J(g)\| = \|g\|_1$. It need not be surjective. In particular, for Lebesgue measure on $[0, 1]$ there exists $T \in (L^\infty([0, 1]))^*$ such that

$$\nexists g \in L^1([0, 1]) \quad \forall f \in L^\infty([0, 1]) : \quad T(f) = \int_0^1 fg \, dm.$$

28 Lecture 28: Vitali's covering theorem

Definition 28.1 (Fine interval cover). Let $E \subseteq \mathbb{R}$ and let $\mathcal{V}$ consist of closed bounded intervals of positive length. The family is a fine cover of E iff

$$\forall x \in E \quad \forall \delta > 0 \quad \exists I \in \mathcal{V} : \quad x \in I, \quad 0 < |I| < \delta.$$

For $I = [a, b]$, $|I| = b - a$ denotes its length. For $J = [c - \ell/2, c + \ell/2]$ and $t > 0$, write $tJ = [c - t\ell/2, c + t\ell/2]$.

Lemma 28.2 (Geometric selection bound). For closed bounded intervals I, J with $|J| > 0$,

$$I \cap J \neq \emptyset, \quad |I| \leq 2|J| \implies I \subseteq 5J.$$

Theorem 28.3 (Vitali covering). *Let $E \subseteq \mathbb{R}$, $m^*(E) < \infty$, and let $\mathcal{V}$ be a fine cover of E . There exists a finite or countably infinite subfamily $(I_j)_{j \in J}$, $J \subseteq \mathbb{N}$, of $\mathcal{V}$ such that*

$$\forall i, j \in J, i \neq j \implies I_i \cap I_j = \emptyset, \quad m^* \left(E \setminus \bigcup_{j \in J} I_j \right) = 0.$$

No measurability assumption on E is needed.

Corollary 28.4 (Localization and finite selections). *The countable-selection conclusion also holds for arbitrary $E \subseteq \mathbb{R}$, without $m^*(E) < \infty$. If E is bounded, then*

$$\forall \varepsilon > 0 \exists r \geq 0 \exists \text{ disjoint } I_1, \dots, I_r \in \mathcal{V} : \quad m^* \left(E \setminus \bigcup_{j=1}^r I_j \right) < \varepsilon.$$

29 Lecture 29: Monotone functions and bounded variation

In Sections 29–30, $a, b \in \mathbb{R}$ satisfy $a < b$ and integrals over $[a, b]$ use the restricted Lebesgue space $([a, b], \{E \cap [a, b] : E \in \mathcal{L}(\mathbb{R})\}, m|_{[a, b]})$.

Definition 29.1 (Monotonicity and differentiability). A function $F : [a, b] \rightarrow \mathbb{R}$ is nondecreasing iff $\forall x, y \in [a, b], x \leq y \implies F(x) \leq F(y)$. It is nonincreasing iff the inequality is reversed, and monotone iff it is one of these types. For $x \in (a, b)$ the derivative exists as a finite real number d iff

$$\forall \varepsilon > 0 \exists \delta > 0 \forall h \in \mathbb{R} : \quad \left[0 < |h| < \delta, x + h \in [a, b] \implies \left| \frac{F(x+h) - F(x)}{h} - d \right| < \varepsilon \right].$$

In that case $F'(x) = d$.

Definition 29.2 (Four Dini derivatives). For $x \in (a, b)$ define, in $\overline{\mathbb{R}}$,

$$\begin{aligned} \overline{D}_+ F(x) &= \limsup_{h \downarrow 0} \frac{F(x+h) - F(x)}{h}, & \underline{D}_+ F(x) &= \liminf_{h \downarrow 0} \frac{F(x+h) - F(x)}{h}, \\ \overline{D}_- F(x) &= \limsup_{h \downarrow 0} \frac{F(x) - F(x-h)}{h}, & \underline{D}_- F(x) &= \liminf_{h \downarrow 0} \frac{F(x) - F(x-h)}{h}. \end{aligned}$$

The limits restrict $h > 0$ to $x \pm h \in [a, b]$. For example,

$$\overline{D}_+ F(x) = \inf_{\delta > 0} \sup_{0 < h < \min\{\delta, b-x\}} \frac{F(x+h) - F(x)}{h},$$

and the corresponding liminf is $\sup_{\delta > 0} \inf_{0 < h < \min\{\delta, b-x\}}$ of that quotient. F is differentiable at x iff all four Dini derivatives agree at a finite value; that value is $F'(x)$.

Theorem 29.3 (Differentiability of monotone functions). *Every finite-valued monotone $F : [a, b] \rightarrow \mathbb{R}$ is differentiable m -a.e. on (a, b) . Define F' to be zero on the null set where the finite derivative does not exist and at the endpoints. This gives a measurable representative. If F is nondecreasing, then*

$$F' \geq 0 \quad m\text{-a.e.}, \quad F' \in L^1([a, b]), \quad \int_a^b F'(x) dx \leq F(b) - F(a).$$

For a monotone function on any open interval, differentiability is local and holds a.e. on that interval.

Definition 29.4 (Total variation and bounded variation). For $F : [a, b] \rightarrow \mathbb{R}$ define

$$\text{Var}_{[a,b]} F = \sup \left\{ \sum_{j=1}^r |F(x_j) - F(x_{j-1})| : r \in \mathbb{N}, a = x_0 < x_1 < \cdots < x_r = b \right\}.$$

The class $BV([a, b])$ consists of all such F with $\text{Var}_{[a,b]} F < \infty$. For $x \in [a, b]$ put $V(x) = \text{Var}_{[a,x]} F$, with $V(a) = 0$.

Theorem 29.5 (Decomposition of a BV function). For $F \in BV([a, b])$ define

$$P(x) = \frac{V(x) + F(x) - F(a)}{2}, \quad N(x) = \frac{V(x) - F(x) + F(a)}{2}.$$

The functions P, N are finite and nondecreasing, $P(a) = N(a) = 0$, and

$$\forall x \in [a, b] : \quad V(x) = P(x) + N(x), \quad F(x) = F(a) + P(x) - N(x).$$

For $a \leq c \leq b$, $\text{Var}_{[a,b]} F = \text{Var}_{[a,c]} F + \text{Var}_{[c,b]} F$.

Corollary 29.6 (Characterization and derivative bound). For $F : [a, b] \rightarrow \mathbb{R}$,

$$F \in BV([a, b]) \iff \exists G, H : [a, b] \rightarrow \mathbb{R} \text{ nondecreasing} : \quad F = G - H.$$

Every BV function is Borel measurable and differentiable a.e., and its derivative representative satisfies

$$F' \in L^1([a, b]), \quad \int_a^b |F'(x)| dx \leq \text{Var}_{[a,b]} F.$$

30 Lecture 30: Absolutely continuous functions

Definition 30.1 (Absolute continuity on an interval). A function $F : [a, b] \rightarrow \mathbb{R}$ belongs to $AC([a, b])$ iff

$$\forall \varepsilon > 0 \exists \delta > 0 \forall r \in \mathbb{N} \forall a \leq a_j < b_j \leq b \ (1 \leq j \leq r) : \\ \left[(a_i, b_i) \cap (a_j, b_j) = \emptyset \ (i \neq j), \sum_{j=1}^r (b_j - a_j) < \delta \implies \sum_{j=1}^r |F(b_j) - F(a_j)| < \varepsilon \right].$$

Thus the choice of δ is independent of the number and location of the intervals.

Theorem 30.2 (Primitive of an integrable function). For every $g \in L^1([a, b])$ and $c \in \mathbb{R}$, the function

$$F(x) = c + \int_a^x g(t) dt \quad (x \in [a, b])$$

belongs to $AC([a, b])$ and satisfies $F' = g$ m-a.e.

Theorem 30.3 (Characterization and fundamental theorem of calculus). For $F : [a, b] \rightarrow \mathbb{R}$,

$$F \in AC([a, b]) \iff \exists g \in L^1([a, b]) \forall x \in [a, b] : \quad F(x) = F(a) + \int_a^x g(t) dt.$$

Such g is unique up to a.e. equality and equals F' a.e. In particular,

$$F \in AC([a, b]) \implies F \in BV([a, b]), \quad F' \in L^1([a, b]), \quad \forall x \in [a, b], \quad F(x) = F(a) + \int_a^x F'(t) dt.$$

Every absolutely continuous function is uniformly continuous:

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in [a, b] : \quad |x - y| < \delta \implies |F(x) - F(y)| < \varepsilon.$$

Corollary 30.4 (Variation and zero derivative). If $F \in AC([a, b])$, then

$$\text{Var}_{[a,b]} F = \int_a^b |F'(t)| dt, \quad F' = 0 \text{ m-a.e.} \iff \forall x \in [a, b], \quad F(x) = F(a).$$

31 Lecture 31: Stieltjes measures and distribution functions

Definition 31.1 (Local finiteness and normalized cumulative function). A Borel measure μ on $\mathbb{R}$ is locally finite iff

$$\forall a, b \in \mathbb{R}, a < b, \quad \mu([a, b]) < \infty.$$

On $\mathbb{R}$ this is equivalent to finite measure on every compact set, or to every point having an open neighborhood of finite measure. Every locally finite Borel measure is sigma-finite. Its cumulative function normalized at 0 is

$$G_\mu(x) = \begin{cases} \mu((0, x]), & x \geq 0, \\ -\mu((x, 0]), & x < 0. \end{cases}$$

Definition 31.2 (Right continuity and left limit). A function $F : \mathbb{R} \rightarrow \mathbb{R}$ is right-continuous iff

$$\forall x \in \mathbb{R} \forall \varepsilon > 0 \exists \delta > 0 \forall y \in [x, x + \delta) : |F(y) - F(x)| < \varepsilon.$$

For finite nondecreasing F , $F(x-) = \lim_{y \uparrow x} F(y) = \sup_{y < x} F(y)$ is finite.

Theorem 31.3 (Lebesgue–Stieltjes correspondence). *For every finite-valued right-continuous nondecreasing $F : \mathbb{R} \rightarrow \mathbb{R}$, there is a unique locally finite Borel measure μ_F such that*

$$\forall a, b \in \mathbb{R}, a < b : \mu_F((a, b]) = F(b) - F(a).$$

Conversely, G_μ is finite-valued, right-continuous, and nondecreasing for every locally finite μ , and $\mu_{G_\mu} = \mu$. For any two such functions,

$$\mu_F = \mu_G \iff \exists c \in \mathbb{R} \forall x \in \mathbb{R}, F(x) - G(x) = c.$$

In particular, $G_{\mu_F}(x) = F(x) - F(0)$.

Example 31.4 (Sigma-finiteness alone is insufficient). Define the Borel density $h(x) = |x|^{-1}$ for $x \neq 0$ and $h(0) = 0$, and set $\mu(A) = \int_A h \, dm$ for $A \in \mathcal{B}(\mathbb{R})$. The sets $\{0\}$ and $\{x : 1/n \leq |x| \leq n\}$, $n \in \mathbb{N}$, form a countable finite-measure cover of $\mathbb{R}$. Nevertheless, $\forall \varepsilon > 0, \mu([-\varepsilon, \varepsilon]) = \infty$. Thus μ is sigma-finite but not locally finite.

Definition 31.5 (Probability distribution function). A cumulative distribution function is an $F : \mathbb{R} \rightarrow [0, 1]$ which is nondecreasing, right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1.$$

For a Borel probability measure μ its distribution function is $F_\mu(x) = \mu((-\infty, x])$.

Theorem 31.6 (Probability correspondence). *The maps $\mu \mapsto F_\mu$ and $F \mapsto \mu_F$ are mutually inverse bijections between Borel probability measures on $\mathbb{R}$ and cumulative distribution functions.*

Definition 31.7 (Point atoms and atomic measures). Let μ be a locally finite Borel measure on $\mathbb{R}$. A point $x \in \mathbb{R}$ is a point atom of μ iff $\mu(\{x\}) > 0$. Let $D_\mu = \{x : \mu(\{x\}) > 0\}$. A measure is nonatomic in the point sense used here iff $D_\mu = \emptyset$. It is purely atomic iff it is supported on a countable set of point atoms, equivalently iff

$$\mu = \sum_{j \in J} a_j \delta_{x_j}$$

for a finite or countable index set J , distinct x_j , and $a_j > 0$.

Proposition 31.8 (Jumps and the atomic part). *For a locally finite $\mu = \mu_F$,*

$$\forall x \in \mathbb{R}, \quad \mu(\{x\}) = F(x) - F(x-).$$

The set D_μ is countable, and F is continuous iff $D_\mu = \emptyset$. With $a_x = \mu(\{x\})$,

$$\mu_{\text{at}}(A) = \sum_{x \in D_\mu \cap A} a_x, \quad \mu_{\text{na}}(A) = \mu(A \setminus D_\mu)$$

are the unique purely atomic and nonatomic parts, and $\mu = \mu_{\text{at}} + \mu_{\text{na}}$. The two measures are mutually singular.

Definition 31.9 (Locally absolute continuity and singular continuity). A function is locally absolutely continuous on $\mathbb{R}$ iff its restriction to every compact interval $[a, b]$ belongs to $AC([a, b])$. A locally finite measure ρ is singular continuous relative to m iff

$$\rho \perp m, \quad \forall x \in \mathbb{R}, \quad \rho(\{x\}) = 0.$$

For a finite-valued continuous nondecreasing F , the phrase *singular continuous function* means that μ_F is singular continuous.

Theorem 31.10 (Density and derivative of a cumulative function). *For finite-valued right-continuous nondecreasing F , let $\mu_F = h m + \rho$ be its Lebesgue decomposition on $\mathcal{B}(\mathbb{R})$. Then $h \geq 0$, $h \in L^1_{\text{loc}}(\mathbb{R})$, and*

$$F' = h \quad m\text{-a.e.}, \quad L^1_{\text{loc}}(\mathbb{R}) = \{u \text{ measurable} : \forall a < b, \int_a^b |u| dm < \infty\}.$$

In particular,

$$\mu_F \perp m \iff F' = 0 \quad m\text{-a.e.}, \quad \mu_F \ll m \iff F \in AC_{\text{loc}}(\mathbb{R}).$$

If F is continuous, $\mu_F \perp m$ is equivalent to singular continuity as defined above.

Theorem 31.11 (Three-part decomposition of a probability measure). *Every Borel probability μ on $\mathbb{R}$ has a unique decomposition*

$$\mu = \mu_{\text{at}} + \mu_{\text{sc}} + \mu_{\text{ac}}, \quad \mu_{\text{ac}} = h m,$$

where, for a finite or countable set of distinct points,

$$\mu_{\text{at}} = \sum_{j \in J} a_j \delta_{x_j}, \quad a_j > 0, \quad \mu_{\text{sc}} \perp m, \quad \forall x, \quad \mu_{\text{sc}}(\{x\}) = 0, \quad h \geq 0, \quad h \in L^1(m).$$

The density is unique m -a.e.; all three measures are nonnegative and pairwise mutually singular, and

$$\sum_{j \in J} a_j + \mu_{\text{sc}}(\mathbb{R}) + \int_{\mathbb{R}} h dm = 1.$$

If $F = F_\mu$ and $F_i(x) = \mu_i((-\infty, x])$ for $i \in \{\text{at}, \text{sc}, \text{ac}\}$, then for every x ,

$$F(x) = F_{\text{at}}(x) + F_{\text{sc}}(x) + F_{\text{ac}}(x), \quad F_{\text{at}}(x) = \sum_{x_j \leq x} a_j, \quad F_{\text{ac}}(x) = \int_{-\infty}^x h(t) dt.$$

Every component has limit 0 at $-\infty$. F_{sc} is continuous and nondecreasing with derivative zero a.e.; $F'_{\text{at}} = 0$ a.e.; $F' = h$ a.e. These normalization and measure conditions uniquely determine the component functions.

Corollary 31.12 (Local version and additive constants). *For any finite-valued right-continuous nondecreasing $F : \mathbb{R} \rightarrow \mathbb{R}$, decompose the locally finite measure μ_F into $\mu_{\text{at}} + \mu_{\text{sc}} + h \, m$ as above, with $h \in L^1_{\text{loc}}(m)$. Define $G_i = G_{\mu_i}$, so $G_i(0) = 0$. Then*

$$\forall x \in \mathbb{R}, \quad F(x) = F(0) + G_{\text{at}}(x) + G_{\text{sc}}(x) + G_{\text{ac}}(x), \quad G_{\text{ac}}(x) = \int_0^x h(t) \, dt.$$

For $x < 0$, $\int_0^x h = -\int_x^0 h$. The normalized atomic part is

$$G_{\text{at}}(x) = \begin{cases} \sum_{0 < x_j \leq x} a_j, & x \geq 0, \\ -\sum_{x < x_j \leq 0} a_j, & x < 0. \end{cases}$$

Every sum over a bounded interval is finite in total mass. The normalized singular continuous part is continuous with derivative zero a.e. These three functions are unique with this normalization.

32 Lecture 32: The Cantor set and Cantor function

Definition 32.1 (Cantor ternary set). Let $K_0 = [0, 1]$ and recursively define

$$K_{n+1} = \frac{1}{3}K_n \cup \left(\frac{2}{3} + \frac{1}{3}K_n\right) \quad (n \in \{0, 1, 2, \dots\}), \quad K = \bigcap_{n=0}^{\infty} K_n.$$

Here $cE + t = \{cx + t : x \in E\}$. Equivalently, K_{n+1} results from deleting the open middle third of each interval component of K_n .

Proposition 32.2 (Structure, cardinality, and measure). *K_n is the union of 2^n disjoint closed intervals of length 3^{-n} , and*

$$m(K_n) = (2/3)^n, \quad m(K) = 0.$$

The set K is compact, perfect, and nowhere dense. Explicitly,

$$\forall x \in K \, \forall r > 0 : \quad ((x - r, x + r) \setminus \{x\}) \cap K \neq \emptyset, \quad \text{int}(\overline{K}) = \emptyset.$$

It has cardinality $|K| = |\mathbb{R}| = 2^{\aleph_0}$, and

$$K = \left\{ \sum_{n=1}^{\infty} \frac{a_n}{3^n} : (a_n) \in \{0, 2\}^{\mathbb{N}} \right\}.$$

Thus membership means the existence of a ternary expansion using only 0, 2; it does not require every ternary expansion to have that property. In particular, $1 = \sum_{n \geq 1} 2/3^n$ is included.

Definition 32.3 (Cantor function). For $x \in K$ use its $\{0, 2\}$ -digit expansion and set

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3^n} \implies C(x) = \sum_{n=1}^{\infty} \frac{a_n/2}{2^n}.$$

For each connected component (u, v) of $[0, 1] \setminus K$ define $C(x) = C(u) = C(v)$ for $x \in (u, v)$. The endpoint values agree and the definition is well defined.

Theorem 32.4 (Properties and functional equations). *The Cantor function $C : [0, 1] \rightarrow [0, 1]$ is continuous and nondecreasing, and*

$$C(0) = 0, \quad C(1) = 1, \quad \text{Var}_{[0,1]} C = 1, \quad C' = 0 \quad m\text{-a.e.}$$

It satisfies

$$C(x) = \begin{cases} \frac{1}{2}C(3x), & 0 \leq x \leq \frac{1}{3}, \\ \frac{1}{2}, & \frac{1}{3} \leq x \leq \frac{2}{3}, \\ \frac{1}{2} + \frac{1}{2}C(3x - 2), & \frac{2}{3} \leq x \leq 1. \end{cases}$$

It is Hölder continuous with exponent $\alpha = \log 2 / \log 3$:

$$\exists H < \infty \forall x, y \in [0, 1] : |C(x) - C(y)| \leq H|x - y|^\alpha.$$

Corollary 32.5 (Singular continuous probability). *Extend C to $\mathbb{R}$ by $C(x) = 0$ for $x < 0$ and $C(x) = 1$ for $x > 1$. Its Stieltjes measure is a Borel probability satisfying*

$$\mu_C(K) = 1, \quad \forall x \in \mathbb{R}, \mu_C(\{x\}) = 0, \quad \mu_C \perp m.$$

Consequently C is singular continuous and $C|_{[0,1]} \notin AC([0, 1])$, with

$$\int_0^1 C'(x) dx = 0 < 1 = C(1) - C(0).$$

Corollary 32.6 (Lebesgue sets beyond the Borel sigma-algebra).

$$\forall E \subseteq K, \quad E \in \mathcal{L}(\mathbb{R}), \quad m(E) = 0, \quad \exists E \subseteq K : E \notin \mathcal{B}(\mathbb{R}).$$

Thus $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R})$.

Convergence implications with their hypotheses

All functions below have real measurable representatives. The expressions *a.e.*, *almost uniformly*, and *in measure* have exactly the quantified definitions in Sections 22–23. In the L^p rows $1 \leq p < \infty$.

Hypothesis	Conclusion
$f_n \rightarrow f$ uniformly outside a single measurable null set	$f_n \rightarrow f$ almost uniformly.
$f_n \rightarrow f$ almost uniformly	$f_n \rightarrow f$ a.e. and $f_n \xrightarrow{\mu} f$.
$\mu(\Omega) < \infty$, $f_n \rightarrow f$ a.e.	$f_n \rightarrow f$ almost uniformly and in measure.
$f_n \xrightarrow{\mu} f$	$\exists n_1 < n_2 < \dots$ with $f_{n_k} \rightarrow f$ a.e.
$f_n, f \in L^p(\mu)$, $\ f_n - f\ _p \rightarrow 0$	$f_n \xrightarrow{\mu} f$, a subsequence converges a.e., and $\ f_n\ _p \rightarrow \ f\ _p$.
$f_n \in L^p(\mu)$, $f_n \xrightarrow{\mu} f$, and $\nu_n = f_n ^p \mu$ equicontinuous at $\emptyset$	$f \in L^p(\mu)$ and $\ f_n - f\ _p \rightarrow 0$.
$f_n \rightarrow f$ a.e., $ f_n \leq g$ a.e. for every n , $g \in L^1(\mu)$, $g \geq 0$	$f_n, f \in L^1(\mu)$, $\ f_n - f\ _1 \rightarrow 0$, and $\int f_n d\mu \rightarrow \int f d\mu$.

Sources and scope

- [1] Claudio Landim, official course page: <https://w3.impa.br/~landim/lectures.html>.
- [2] Official IMPA YouTube playlist, 32 main lectures and three supplementary exercise-solution videos: [Claudio Landim – Measure Theory playlist](#).
- [3] Complete annotated course guide hosted by IMPA, dated November 21, 2022: <https://w3.impa.br/~lacoïn/measurecomments2022.pdf>.
- [4] IMPA teaching page linking the course and annotated guide: <https://w3.impa.br/~lacoïn/teaching.html>.

The mathematical statements are independently formulated. This is a rigorous reformulation of the preceding reference, rather than a verbatim transcription or a claim that every sentence on the lecture blackboard has been catalogued.